

ARM PrimeCell

Vectored Interrupt Controller (PL190)

Release Note

ARM PrimeCell Vectored Interrupt Controller (PL190) Release Note

© Copyright ARM Limited 2000. All rights reserved.

Proprietary notice

ARM, the ARM powered logo, Thumb and StrongARM are registered trademarks of ARM Limited.

The ARM logo, PrimeCell, AMBA, Angel, ARMulator, EmbeddedICE, ModelGen, Multi-ICE, ARM7TDMI, ARM9TDMI, TDMI and STRONG are trademarks of ARM Limited.

All other products or services mentioned herein may be trademarks of their respective owners.

Neither the whole nor any part of the information contained in, or the product described in, this document may be adapted or reproduced in any material form except with the prior written permission of the copyright holder.

The product described in this document is subject to continuous developments and improvements. All particulars of the product and its use contained in this document are given by ARM Limited in good faith. However, all warranties implied or expressed, including but not limited to implied warranties or merchantability, or fitness for purpose, are excluded.

This document is intended only to assist the reader in the use of the product. ARM Limited shall not be liable for any loss or damage arising from the use of any information in this document, or any error or omission in such information, or any incorrect use of the product.

Document confidentiality status

This document is Open Access. This document has no restriction on distribution.

Product status

The information in this document is Final (information on a developed product).

Web address

<http://www.arm.com>

Feedback

ARM limited welcomes feedback on both the product, and the documentation.

Feedback on this document

If you have any comments on about this document, please send email to errata@arm.com giving:

- The document title
- The documents number
- The page number(s) to which your comments refer
- A concise explanation of your comments

General suggestion for additions and improvements are also welcome.

Feedback on the ARM PrimeCell Multimedia Card Interface

If you have any comments or suggestions about this product, please contact your supplier giving:

- The Product name
- A concise explanation of your comments.

Change history

Issue	Date	Change
A	31 st August 2000	First release.
B	30 th October 2000	Second release.

Contents

1	PRODUCT DELIVERABLES	1
1.1	ARM Part Numbers for this product	1
1.2	File deliverables	2
2	INSTALLATION	3
2.1	Introduction	3
2.2	Disk space required	3
2.3	Installation procedure	3
2.3.1	Unpacking the shipment	3
2.3.2	Re-creating the peripheral directory structure	4
2.3.3	Installation of the reference cell library	5
2.3.4	Installation of the Generic User Guide	5
3	DOCUMENTATION	6
3.1	Adobe PDF	6
3.2	Peripheral Documents Supplied	6
4	TRIAL TESTING	7
4.1	Product Maturity	7
4.2	Technical data summary	7
4.3	BusTalk Functional block verification test scenarios	8
4.3.1	Synchronous clocks	8
5	DIFFERENCES FROM PREVIOUS REVISIONS	8
5.1	Release 1.1 (REL1v1)	8
5.1.1	RTL correction to avoid meta-stability issue	8
6	KNOWN ISSUES	8
7	REVISION HISTORY	8

1 PRODUCT DELIVERABLES

1.1 ARM Part Numbers for this product

Table 1.1-1 lists the ARM part numbers for the individual deliverables that form part of the delivery for this ARM PrimeCell peripheral:

Product deliverables for the ARM PrimeCell PL190 peripheral	Document number	Product code
Synthesizable VHDL RTL	N/A	PL190-MN-23001-REL1v1
Synthesizable Verilog RTL	N/A	PL190-MN-22001-REL1v1
Synthesis Scripts (Synopsys Design Compiler, Avant! CB25 cell library)	N/A	PL190-MN-01002-REL1v1
Testbench - Functional Verification, VHDL	N/A	PL190-MN-23010-REL1v1
Testbench - EASY Integration test, VHDL	N/A	PL190-MN-23012-REL1v1
Testbench - Functional Verification, Verilog	N/A	PL190-MN-22010-REL1v1
Testbench - EASY Integration test, Verilog	N/A	PL190-MN-22012-REL1v1
Test vectors-BusTalk, functional verification	N/A	PL190-VE-01010-REL1v1
Test vectors-BusTalk, integration test	N/A	PL190-VE-01012-REL1v1
ARM PrimeCell Vectored Interrupt Controller (PL190) Technical Reference Manual (PDF)	ARM DDI 0181 C	PL190-DA-03001-REL1v1
ARM PrimeCell Vectored Interrupt Controller (PL190) Design Manual (PDF)	PL190 DDES 0000 A	PL190-DC-02002-REL1v1
ARM PrimeCell Vectored Interrupt Controller (PL190) Integration Manual (PDF)	PL190 INTM 0000 A	PL190-DC-10002-REL1v1
ARM PrimeCell Vectored Interrupt Controller (PL190) Release Note (PDF)	PL190 RELN 0000 A	PL190-DC-06002-REL1v1

Table 1.1-1: ARM part numbers for ARM PrimeCell VIC PL190

Note Only the required deliverables will be supplied, as per the contractual agreement.

For each of the above, two files are generated using the UNIX tar create and the non-document tar files are then UNIX compressed. The files are named for each unique product revision, and the generic naming convention is illustrated below :

PL190-AA-NNNN-REL1v1.tar.Z - compressed tar file

PL190-AA-NNNN-REL1v1.lst - directory and file listing of above tar file

Where -AA-NNNN represents the alpha-numeric product sub-type,
and REL1v1 represents the numeric release version of this peripheral.

1.2 File deliverables

This ARM PrimeCell peripheral consists of the following compressed 'tar' and 'list' file deliverables:

PLXXX_PrimeCell/PL190/REL1v1/PL190-DC-06002-REL1v1.pdf

PLXXX_PrimeCell/PL190/REL1v1/DAtasheet/PL190-DA-03001-REL1v1/PL190-DA-03001-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/DAtasheet/PL190-DA-03001-REL1v1/PL190-DA-03001-REL1v1.tar

PLXXX_PrimeCell/PL190/REL1v1/DoCumentation/PL190-DC-02002-REL1v1/PL190-DC-02002-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/DoCumentation/PL190-DC-02002-REL1v1/PL190-DC-02002-REL1v1.tar

PLXXX_PrimeCell/PL190/REL1v1/DoCumentation/PL190-DC-10002-REL1v1/PL190-DC-10002-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/DoCumentation/PL190-DC-10002-REL1v1/PL190-DC-10002-REL1v1.tar

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-01002-REL1v1/PL190-MN-01002-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-01002-REL1v1/PL190-MN-01002-REL1v1.tar.Z

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-22001-REL1v1/PL190-MN-22001-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-22001-REL1v1/PL190-MN-22001-REL1v1.tar.Z

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-22010-REL1v1/PL190-MN-22010-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-22010-REL1v1/PL190-MN-22010-REL1v1.tar.Z

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-22012-REL1v1/PL190-MN-22012-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-22012-REL1v1/PL190-MN-22012-REL1v1.tar.Z

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-23001-REL1v1/PL190-MN-23001-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-23001-REL1v1/PL190-MN-23001-REL1v1.tar.Z

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-23010-REL1v1/PL190-MN-23010-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-23010-REL1v1/PL190-MN-23010-REL1v1.tar.Z

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-23012-REL1v1/PL190-MN-23012-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/MN_native_model/PL190-MN-23012-REL1v1/PL190-MN-23012-REL1v1.tar.Z

PLXXX_PrimeCell/PL190/REL1v1/VEctors/PL190-VE-01010-REL1v1/PL190-VE-01010-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/VEctors/PL190-VE-01010-REL1v1/PL190-VE-01010-REL1v1.tar.Z

PLXXX_PrimeCell/PL190/REL1v1/VEctors/PL190-VE-01012-REL1v1/PL190-VE-01012-REL1v1.lst

PLXXX_PrimeCell/PL190/REL1v1/VEctors/PL190-VE-01012-REL1v1/PL190-VE-01012-REL1v1.tar.Z

A generic peripheral user guide document (PDF) is supplied. This describes instructions for synthesis and simulation.

PLXXX_PrimeCell/PL000_generic/DoCumentation/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0.lst

PLXXX_PrimeCell/PL000_generic/DoCumentation/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0.tar

An install script is supplied which uncompresses and tar extracts (un-tar) the compressed tar files in order to build the peripheral directory structure.

PLXXX_PrimeCell/PL000_generic/SoftWare/PL000-SW-99000-REL1v0/INSTALL

Note It is possible that the revision number (used in directory paths) for the INSTALL script and user guide may have been updated beyond REL1v0 described above.

A reference cell library (Avant! CB25 0.25µm standard cells) is supplied with this peripheral.

PLXXX_PrimeCell/PL000_generic/Avanti/CB25_1999.1.tar.Z

PLXXX_PrimeCell/PL000_generic/Avanti/CB25_1999.1.lst

2 INSTALLATION

2.1 Introduction

Intellectual Property (IP) deliverables are collectively delivered as a single UNIX compressed tar file which has been security encrypted.

An ARM PrimeCell peripheral may be one of several products delivered as part of this single encrypted file. Each ARM PrimeCell peripheral comprises a set of compressed tar files which correspond to a unique part number quoted in the contractual agreement document. A corresponding list file details the files in each tar file

These installation instructions cover the following platform/operating system:

- UNIX

2.2 Disk space required

This installation will require 30 Mbytes of free disk space.

2.3 Installation procedure

The installation procedure is summarised below:

2.3.1 Unpacking the shipment

The following steps describe how to unpack the separate products delivered in this shipment.

1. Relocate the shipment file

Relocate the encrypted file (for example <file>.tar.Z.des, if DES encryption has been used) to an appropriate location that has sufficient free disk space.

2. Decrypt file

Follow the accompanying instructions for decryption of the file supplied, to generate a decrypted, compressed, tar file <file>.tar.Z.

3. Uncompress file

Execute the UNIX uncompress command to obtain the tar file <file>.tar:

```
uncompress <file>.tar.Z
```

where <file> is the name of the file supplied.

Note Invoke `gzip -d <file>.tar.gz`, if gunzip compression has been applied.

4. Extract tar files

To extract individual product tar files, execute the UNIX tar extract command:

```
tar -xvf <file>.tar
```

This will generate individual tar and list files for all of the products within this single shipment. Each peripheral consists of a set of tar files, and each tar file is named with the ARM part number that is listed in the contractual agreement document.

Note It is suggested that part numbers delivered are checked against the contractual agreement document.

2.3.2 Re-creating the peripheral directory structure

The following steps describe how to re-create the directory structure for each separate peripheral, which contains HDL RTL files, synthesis scripts, testbenches and documentation. These steps will need to be repeated for other peripherals in this shipment.

5. Choose method of installation

The directory structure can be extracted using the INSTALL script provided, or manually moving the files to the destination directory and using UNIX uncompress and tar commands. The INSTALL script moves the compressed tar files delivered, to a user specified destination directory and then uncompresses and tar extracts the files to re-create the peripheral directory structure. Manual installation may be preferred if only some of the deliverables are required.

Go to step 6 to use the INSTALL script provided, otherwise go to step 7 for manual installation.

All files begin with PL190, where 190 is a 3-digit value unique to each peripheral.

Note It is important to un-install or rename any version of this peripheral that already exists in the destination directory prior to uncompressing and extracting these tar files.

6. Run the install script

The install script supplied will automatically uncompress and tar extract files for the specified peripheral revision, in a user specified directory. When step 6 is complete there is then no need to perform steps 7 and 8, instead go to step 9.

Change to the directory containing the INSTALL script, for example (note, revision may be later than - REL1v0)

```
cd PLXXX_PrimeCell/PL000_generic/SoftWare/PL000-SW-99000-REL1v0
```

To invoke the install C-shell script supplied, enter the following at the UNIX command line:

```
INSTALL <PRODUCT_NUMBER> <REVISION> <DESTINATION_DIRECTORY>
```

Options

<PRODUCT_NUMBER> = Each peripheral is defined by unique product code using the characters PL + 3-numeric digits. See below the shipment directory PLXXX_PrimeCell/

<REVISION> = Each peripheral has a revision number which has a status label and numeric version. See two directories below the shipment directory PLXXX_PrimeCell/

<DESTINATION_DIRECTORY> = the full directory path where the peripheral is to be installed.

For example

A product shipped with the directory structure /PLXXX_PrimeCell/PL190/REL1v1/ has a product code PL190 and revision number Release REL1v1.

To install this peripheral to the directory /ARM/PrimeCell_peripherals enter the following command line

```
INSTALL PL190 REL1v1 /ARM/PrimeCell_peripherals
```

This will install the peripheral having a product number PL190, with revision number REL1v1, to the destination directory /ARM/PrimeCell_peripherals/

Go to step 9

7. Uncompress files

Copy all the compressed 'tar' files (*.tar.Z) for the peripheral, to the chosen destination directory. In the example below, the destination directory is called /ARM/PrimeCell_peripherals/.

Use the UNIX uncompress command to obtain the tar files <file>.tar (* wildcards could also be used):

```
uncompress <file>.tar.Z
```

8. Extract tar files

To extract each file individually, execute the UNIX tar extract command to build the peripheral file structure:

```
tar -xvf <file>.tar
```

9. Peripheral directory installed

When all the peripheral files have been extracted, the destination directory (/ARM/PrimeCell_peripherals/) will contain:

- the top-level peripheral directory (e.g. vic_pl190) and underlying file structure of the deliverables, as well as the Release Note document file (PDF).
- The remaining documentation for the peripheral (in Adobe Portable Document Format, *.pdf), is located in the vic_pl190/docs/ directory.

2.3.3 Installation of the reference cell library

The reference cell library should be copied manually and tar extracted in the directory containing the PrimeCell peripherals. An example is shown below to copy the Avant! CB25 1999.1 cell library to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PLXXX_PrimeCell/PL000_generic/Avanti/CB25_1999.1.tar.Z .
```

Execute the UNIX uncompress and tar extract commands as below:

```
uncompress CB25_1999.1.tar.Z
tar -xvf CB25_1999.1.tar
```

2.3.4 Installation of the Generic User Guide

For instructions on performing simulations and synthesis, refer to the generic peripheral user guide.

The most recent revision of the generic user guide should be copied manually to a suitable location. An example is shown below to copy the generic user guide (revision REL1v0) to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp /PLXXX_PrimeCell/PL000_generic/DoCumentation/PL000-DC-08001-REL1v0/PL000-DC-08001-REL1v0.tar .
```

To extract the document file (PDF), execute the UNIX tar extract command:

```
tar -xvf PL000-DC-08001-REL1v0.tar
```

3 DOCUMENTATION

3.1 Adobe PDF

Documents are supplied as “Adobe PDF” (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using the appropriate file reader. A suitable file reader can be downloaded from the Adobe World Wide Web site at <http://www.adobe.com>. Select “Acrobat Reader” and download the reader for your computer platform/operating system.

3.2 Peripheral Documents Supplied

The following documents are supplied for the ARM PrimeCell Multimedia Card Interface PL190:
The document files are located in the vic_pl190/docs/ directory.

ARM PrimeCell Vectored Interrupt Controller PL190 Technical Reference Manual ARM DDI 0181

The technical reference manual provides details for programming and interfacing to this ARM PrimeCell peripheral block.

ARM PrimeCell Vectored Interrupt Controller PL190 Integration Manual PL190 INTM 0000

The integration manual provides information required to integrate this ARM PrimeCell block within a typical industry standard ASIC design flow.

ARM PrimeCell Vectored Interrupt Controller PL190 Design Manual PL190 DDES 0000

The design manual describes the implementation details of this ARM PrimeCell block and the testbenches and test programs used for functional verification.

4 TRIAL TESTING

4.1 Product Maturity

This product is newly developed IP and approved for final release.

At the time of release REL1v1, it has not been implemented in production silicon but has been proven on FPGA.

4.2 Technical data summary

The following is a summary of the key technical points from testing of the ARM Vectored Interrupt Controller PL190.

	VHDL RTL	Verilog RTL
Cell library	Avant! Passport 1999.1 CB25 v2.1 (0.25 μm)	Avant! Passport 1999.1 CB25 v2.1 (0.25 μm)
Total cell area, without scan, NAND-2 equivalents	12909	12800
Net interconnect area, without scan, NAND-2 equivalents	6759	6756
Total cell area, scan inserted, NAND-2 equivalents	16291	16168
Net interconnect area, scan inserted, NAND-2 equivalents	7879	7874
Clock frequency used	100MHz	100MHz
Scan insertion figure % (number of 'D' flip-flops replaced)	100% (1050)	100% (1050)
Estimated scan fault coverage, collapsed.	100%	100%
Synthesis and scan test tools and version	Synopsys Design Compiler and Test Compiler 2000.05	Synopsys Design Compiler and Test Compiler 2000.05
Simulation tools and versions	Model Technology Inc, ModelSim v5.3d	Model Technology Inc, ModelSim v5.3d Cadence LDV 3.0
Global Peripheral Development Environment Version	Release REL1v1	

4.3 BusTalk Functional block verification test scenarios

Functional block verification has been performed with the following scenarios using the BusTalk testbench with free-running HCLK.

4.3.1 Synchronous clocks

HCLK	Simulations
100MHz (10 ns)	VHDL and Verilog, RTL and gate-level min/typ/max SDF

For further details of functional verification testing, refer to the ARM PrimeCell Vectored Interrupt Controller (PL190) Design Manual.

To change clock periods modify following line in the verification/\$HDL_SOURCE/tbench/timing.vhd (or .v) file:

```
-- Define the low phase of clock
constant Tclk1      : time    := 5 ns;
-- Define the high phase of clock
constant Tclkh      : time    := 5 ns;
```

or in case of Verilog testbench:

```
// Clock frequency
`define Tclk1    5
`define Tclkh    5
```

5 DIFFERENCES FROM PREVIOUS REVISIONS

5.1 Release 1.1 (REL1v1)

5.1.1 RTL correction to avoid meta-stability issue

It was found that asynchronous interrupts could cause meta-stability problems when reading the Vector Address register. This has been fixed in Release 1.1.

6 KNOWN ISSUES

No known issues.

7 REVISION HISTORY

Date	Revision	Description
31 st August 2000	REL1v0	Initial Release 1.0
30 th October 2000	REL1v1	Release 1.1